

Intelligent Narmada Basin Monitoring & Governance (INBMG)

An AI-Driven Framework for Ecological Conservation and Public Safety

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Executive Summary

Effective management of river basins such as the Narmada requires a transition from reactive response systems to predictive, data-driven decision-making. You need to use data to act before things get out of control. The Narmada River, around Omkareshwar is a place where many people gather and this creates a lot of pressure on the environment and safety.

We created the INBMG framework to tackle these problems. The INBMG framework is a system that uses Computer Vision, Remote Sensing and environmental monitoring to keep track of what's happening in the Narmada Basin. A key innovation of the system is the integration of Environmental DNA (eDNA)-based monitoring for real-time biodiversity assessment. The idea is to connect what the government wants to do with the tools that can make a difference on the ground.

1 Problem Statement & Objectives

The traditional way of managing river basins in India has some problems. Data is fragmented and unstructured and when there is an emergency it takes a long time to respond. Some of the challenges are:

- **Crowd Safety:** The places where people gather for pilgrimages get very crowded and this can lead to stampedes or drowning.
- **Shifting Topography:** The sandbars in the river move and the water currents change, making it dangerous for boatmen and visitors to navigate.
- **Ecological Degradation:** Right now tracking the loss of biodiversity and pollution is a slow and manual process.

Core Objectives: Faster emergency responses by using AI-powered heatmaps, a rock-solid baseline for water quality (pH, contaminants & DNA), and one digital hub for everything, making life easier for both officials and stakeholders.

2 Proposed Solution: The INBMG Architecture

The proposed system introduces a three-part system, custom-built for the Narmada Basin's unique landscape and packed neighborhoods.

2.1 Intelligent Urban Infrastructure & Active Safety

Surveillance cameras on their own aren't enough. Our setup goes further—it actually steps in when things get dangerous.

- **Predictive Crowd Management:** We tap into the CCTV network at key Ghats, running YOLOv8 and SSD-Mobilenet to watch crowd density in real time. If the system spots a risky situation, it alerts local authorities instantly, quick enough to head off trouble before it snowballs.
- **Active Intervention:** If panic strikes or someone's at risk, our drone can drop life jackets automatically, right where they're needed.
- **Smart Navigation Tool:** Boat operators get a navigation that predicts river currents and flags hazards. It keeps boats from colliding, issues voice alerts in English, Hindi, and Ni-madi, and helps everyone stay safe, no one left out.

2.2 Environmental Monitoring & River Health

We combine macro-level satellite surveillance with micro-level biological data.

- **Remote Sensing Mapping:** By analyzing data from Sentinel-1 and LISS-IV using Random Forest and CNN models, the system tracks flood cycles, changing river edges, and illegal encroachment on the floodplains.
- **Micro-Ecological Auditing:** Adapting the CAMP group's methods, we match satellite reflectance with eDNA from water. That means we can track pollution and check on endangered species, all in real time and with a level of detail never seen before.

2.3 Digital Governance & Data Accessibility

Data needs to drive decisions. So we've built a web-based command center for that.

- **The "Pragyambu" Archive:** District officials can use an AI search tool to instantly dig up old technical reports, SAC docs, and past meeting notes.
- **Live Stakeholder Dashboard:** There's layered access for different roles, so everyone—from admins to local workers can see current Water Quality Indices(WQI) and Ghat safety stats at a glance.

3 Strategic Implementation Roadmap

We plan to keep costs reasonable and make change stick, rolling out in three steps:

1. **Phase I (Safety Excellence):** Start in Omkareshwar, launching computer vision crowd alerts and the navigation app.
2. **Phase II (Ecological Resilience):** Then, bring in satellite mapping and DNA-based pollution tracking.

3. **Phase III (Governance Integration):** Lastly, connect every piece to the Digital Dashboard and command center, making live data flow where it matters.

4 Expected Strategic Outcomes

INBMG aligns with the Namami Gange Programm and National Green Tribunal (NGT's) guidance offering a scaled data driven approach for resource management disaster approach. It enables a shift from reactive to predictive action helping the Narmada basin, a bench mark for intelligent river management.